



M1.7 Unit Overview

Describing Data

Students learn to distinguish between categorical and **quantitative data**. They learn to organize **categorical data** in **two-way frequency tables** and create **conditional relative-frequency tables** to make decisions. They analyze one- and two-variable data sets. They use **statistics** appropriate to the shape of the data to compare one-variable data sets (mean and **standard deviation** or median and IQR). They use **scatter plots** with the **line of best fit** and the **correlation coefficients** for two-variable data. They also consider whether the relationship represents **causation** or **correlation**.

Let us know what you thought about this lesson by filling out [this survey](#).

Big Ideas: CC1: Comparing Models, Correlation and Causation, Modeling with Functions, Variability

Explore: A Statistical Question

(Optional) Launch the unit with a non-routine task to investigate the question, "What is a statistical question?"

Investigate: Exploring Climate Change

(Optional) Invite students to investigate a global issue and identify actions to reduce their carbon footprint. Use anytime in this course after Unit 7, Lesson 19.

Sub-Unit 1: Representing Categorical Data (Lessons 1–3)

- Represent categorical data in two-way frequency tables and conditional relative frequency tables.
- Language Goal:** Compare and contrast the types of data that can be gathered from questions on a survey.

Sub-Unit 2: Summarizing One-Variable Data (Lessons 4–12 + Practice Day + Quiz)

- Represent data with dot plots, histograms, and box plots.
- Describe the shape, center, and spread of data.
- Language Goal:** Compare and contrast data sets using measures of center and spread appropriate to the shape of the data set and interpret the differences in context.

Sub-Unit 3: Summarizing Two-Variable Data (Lessons 13–19 + Practice Day)

- Fit a line to data, assess the fit using residual plots, and use the linear model to make predictions.
- Language Goal:** Describe linear associations in data using the line of fit and correlation coefficient.

Review Vocabulary

bin (of a histogram), dot plot, histogram, interquartile range (IQR), linear, linear association, line of fit, maximum, mean, mean absolute deviation (MAD), median, minimum, negative association, non-linear, numerical data, positive association, quartile, range, shape (of a data set), slope, slope-intercept form, y-intercept

Contextual Vocabulary

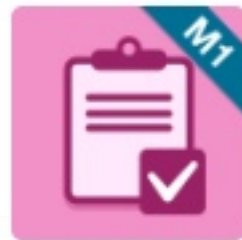
vital information, hearing loss, impact, grant, fish tank, heat wave, minimum wage, income, tree cover, fuel economy, breeding pairs, species, headlines, commute time, green space

Standards

Addressing F-IF.4 F-LE.5 N-Q.2 S-ID.1 S-ID.2 S-ID.3 S-ID.5 S-ID.6 S-ID.6a S-ID.6b S-ID.6c S-ID.7 S-ID.8 S-ID.9 SMP.1 SMP.2
SMP.3 SMP.4 SMP.5 SMP.6 SMP.7 SMP.8

Vocabulary

- association
- bimodal
- bell-shaped
- box plot
- categorical data
- causation
- conditional relative frequency table
- correlation
- correlation coefficient
- joint relative frequency
- line of best fit
- marginal relative frequency
- measure of center
- measure of spread
- outlier
- quantitative data
- r-value
- residual
- residual plot
- scatter plot
- skewed
- standard deviation
- statistic
- symmetric
- two-way frequency table
- uniform



M1.7 Pre-Unit Check



Learn more about your students' understanding of foundational concepts and skills that will support them in the upcoming unit. This pre-unit check is intended for students to complete using a graphing calculator.

OPTIONAL This is optional and can be given in its entirety before the unit begins or spread out throughout the unit.

Let us know what you thought about this lesson by filling out [this survey](#).

Instruction

Differentiation Beyond the Lesson

Big Ideas: CC1: Variability in Data (Grade 6), The Shapes of Distributions (Grade 6), Data Explorations (Grade 8)

Standards

Assessing 6.SP.2 6.SP.3 6.SP.5.a 6.SP.5.c 8.SP.1 8.SP.2 8.SP.3 SMP.2 SMP.4 SMP.6



Explore: A Statistical Question

What is a statistical question?

OPTIONAL

This lesson is optional because this content extends beyond the scope of the standards.

Let us know what you thought about this lesson by filling out [this survey](#).



Student Edition pages and Presentation Screens support learning in this lesson.

Big Ideas: CC1: Modeling with Functions, Variability

Instructional Routines

- Notice and Wonder

Today's Goals

- **Language Goal:** Describe the features of data representations that are important to examine when making conclusions about data sets. *(Writing, Speaking, and Listening)*
- **Language Goal:** Describe the difference between statistical and non-statistical questions. *(Writing, Speaking, and Listening)*

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with the **Building Math Habits of Mind** part of this Explore to support math language acquisition for your students.

Additional Materials

coloring tools

Standards

Building On 6.SP.1 Addressing SMP.2 SMP.6 Building Toward N-Q.1 S-ID.1 S-ID.5 S-ID.6

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10



Survey Says

Lesson 1: What Kinds of Data Can I Collect?

Let's make sense of the kinds of data that can be collected and write questions to get to know each other better.

OPTIONAL This lesson is optional as it addresses standards from a prior course, supporting students in drawing out prior knowledge.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Challenge Creator supports students in creating and sharing challenges with classmates.

Instruction

Differentiation Beyond the Lesson

Big Idea: CC1: Variability

Today's Goals

- Language Goal:** Explain the difference between *quantitative* and *categorical data*. (*Writing, Speaking, and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Additional Vocabulary

survey

Additional Materials

Student Edition: Activity 2

Standards

Building On 6.SP.1 6.SP.5.b Building Toward S-ID.1 S-ID.5 Assessing SMP.3 SMP.6

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.11 ELD.PI.9-10.12 ELD.PI.9-10.3

Vocabulary

- categorical data
- quantitative data

Review Vocabulary

- numerical data

Instructional Routines

- Notice and Wonder
- Decide and Defend
- Collect and Display (MLR2)



Hear Here

Lesson 2: Two-Way Tables and Relative Frequency Tables

Let's use categorical data to make an argument.

Let us know what you thought about this lesson by filling out [this survey](#).



Student Edition pages and Presentation Screens support learning in this lesson.

Instruction

Differentiation Beyond the Lesson

Big Idea: CC1: Variability

Today's Goals

- Represent categorical data in *two-way frequency tables* and *conditional relative frequency tables*.
- Calculate values in two-way frequency tables and conditional relative frequency tables.
- Language Goal:** Analyze two-way frequency tables for *associations* in data. (*Reading, Writing, and Speaking*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 2 to support math language acquisition for your students.

Additional Vocabulary

grand total

Additional Materials

scientific calculators

Standards

Building On 8.SP.4 Addressing S-ID.5 SMP.3 SMP.4 Building Toward S-ID.5

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.11 ELD.PI.9-10.12 ELD.PI.9-10.3 ELD.PI.9-10.6

Vocabulary

- association
- conditional relative frequency table
- two-way frequency table

Instructional Routines

- Decide and Defend
- Think-Pair-Share
- Co-Craft Questions (MLR5)



School Choice

Lesson 3: Making Decisions With Frequency Tables

Let's use data to make decisions and write arguments.

Let us know what you thought about this lesson by filling out [this survey](#).

  Student Edition pages and Presentation Screens support learning in this lesson.

Instruction Differentiation Beyond the Lesson

Big Idea: CC1: Variability

Today's Goals

- Calculate values in total relative frequency tables by row, column, or total.
- Create total relative frequency tables and two-way frequency tables.
- **Language Goal:** Analyze relative frequency tables to determine if there is an association between two variables. (*Reading, Writing, Speaking, and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 3 to support math language acquisition for your students.

Additional Vocabulary

total relative frequency table

Additional Materials

scientific calculators (as needed)

Standards

Building On 8.SP.4 Addressing S-ID.5 SMP.2 SMP.3 SMP.4

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.11 ELD.PI.9-10.12 ELD.PI.9-10.3

Vocabulary

- joint relative frequency
- marginal relative frequency

Instructional Routines

- Think-Pair-Share
- Collect and Display (MLR2)



Love it or Leave it

Lesson 4: Revisiting Dot Plots and Histograms

Let's make sense of dot plots and histograms as ways to visualize one-variable data.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Students use the Desmos Graphing Calculator to create dot plots and histograms. Also, a digital interaction aggregates student responses creating class data displays.

Instruction

Differentiation Beyond the Lesson

Big Idea: CC1: Variability

Today's Goals

- Use technology to represent data with a dot plot or histogram.
- Language Goal:** Describe the advantages of using a dot plot or a histogram to represent data. (*Writing and Speaking*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 2 to support math language acquisition for your students.

Additional Materials

Unit 7 Calculator Guide (optional)

Standards

Building On 6.SP.4 Addressing S-ID.1 SMP.3 Building Toward S-ID.2

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.11 ELD.PI.9-10.12 ELD.PI.9-10.6 ELD.PI.9-10.8 ELD.PII.9-10.4

Review Vocabulary

- bin (of a histogram)
- dot plot
- histogram

Instructional Routines

- Critique, Correct, Clarify (MLR3)



Better Weather?

Lesson 5: Revisiting Box Plots

Let's use box plots to visualize and compare weather data.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Students use the Desmos Graphing Calculator to make box plots and calculate statistics.

Instruction Differentiation Beyond the Lesson

Big Idea: CC1: Variability

Today's Goals

- Use technology to construct *box plots*.
- **Language Goal:** Compare and contrast two data sets represented by *box plots*. (*Speaking and Listening*)
- **Language Goal:** Interpret the parts of *box plots* that represent real-world data. (*Reading, Writing, Speaking, and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 2 to support math language acquisition for your students.

Additional Materials

Unit 7 Calculator Guide (optional)

Standards

Building On 6.SP.4 Addressing N-Q.2 S-ID.1 SMP.2 SMP.3 SMP.4 Building Toward S-ID.2

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.11 ELD.PI.9-10.12 ELD.PI.9-10.2 ELD.PI.9-10.3

Vocabulary

- box plot
- statistic

Review Vocabulary

- maximum (of a data set)
- minimum (of a data set)
- median
- quartile

Instructional Routines

- Stronger and Clearer Each Time (MLR1)
- Collect and Display (MLR2)
- Think-Pair-Share



Shapes of Data

Lesson 6: Describing Data Sets

Let's describe different shapes of data.

 This activity contains a Polygraph. [Learn how to play.](#)

 Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also needed.

Why digital? The digital version of Polygraph coordinates pairings and question exchanges to support students in playfully developing language to describe different shapes of data sets.

Instruction

Differentiation Beyond the Lesson

Big Idea: CC1: Variability

Today's Goal

- **Language Goal:** Describe the shape of a data set using terms like bimodal, bell-shaped, skewed, symmetric, and uniform. (*Writing, Speaking, and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Additional Materials

Student Edition: Activity 3; Activity 3 Cards (one set per pair, pre-cut)

Standards

Building On 6.SP.2 6.SP.4 Addressing S-ID.1 SMP.3 SMP.6 SMP.7 Building Toward S-ID.2 S-ID.3

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.12 ELD.PI.9-10.3 ELD.PI.9-10.5 ELD.PI.9-10.6

Vocabulary

- bell-shaped
- bimodal
- skewed
- symmetric
- uniform

Review Vocabulary

- shape (of a data set)

Instructional Routines

- Which One Doesn't Belong?
- Collect and Display (MLR2)



Quick Pick

Lesson 7: Revisiting Measures of Center

Let's explore how extreme values impact mean and median.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Students use the Desmos Graphing Calculator to calculate means and medians. They also play a digital game to help them make sense of data sets used in Activity 1.

Instruction

Differentiation Beyond the Lesson

Big Idea: CC1: Variability

Today's Goals

- Use technology to calculate the *measures of center* (mean and median) for a data set.
- **Language Goal:** Explain how to calculate the mean and median, and what these tell us about a data set. (*Speaking and Listening*)
- **Language Goal:** Explain the effect of extreme values on the mean and median. (*Writing, Speaking, and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 2 to support math language acquisition for your students.

Additional Materials

Unit 7 Calculator Guide (optional)

Standards

Building On 6.SP.3 6.SP.5.c S-ID.1 Addressing S-ID.2 S-ID.3 SMP.3 SMP.8 Building Toward S-ID.2

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.11 ELD.PI.9-10.12 ELD.PI.9-10.6

Vocabulary

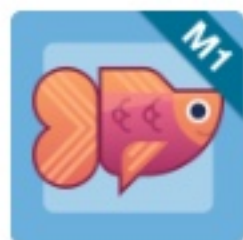
- measure of center

Review Vocabulary

- mean
- median

Instructional Routines

- Notice and Wonder
- Collect and Display (MLR2)
- Decide and Defend



Finding Desmo

Lesson 8: Introduction to Standard Deviation

Let's explore what standard deviation describes about a data set.

📄 Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Students use digital interactions to develop conceptual understanding of standard deviation and calculate it.

Instruction

Differentiation Beyond the Lesson

Big Idea: CC1: Variability

Today's Goals

- Understand that *standard deviation* is a *measure of spread* and use technology to calculate the standard deviation for a data set.
- Language Goal:** Compare and contrast data sets using standard deviation. (*Speaking and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Additional Materials

Unit 7 Calculator Guide (optional)

Standards

Building On 6.SP.3 6.SP.5.c 7.SP.3 Addressing S-ID.2 SMP.2 Building Toward S-ID.3

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.12 ELD.PI.9-10.2

Vocabulary

- measure of spread
- standard deviation

Review Vocabulary

- interquartile range (IQR)
- MAD
- range

Instructional Routines

- Collect and Display (MLR2)
- Think-Pair-Share



Wavering Weather

Lesson 9: Comparing Data Using Mean and Standard Deviation

Let's use mean and standard deviation to compare the temperatures in different cities.

Let us know what you thought about this lesson by filling out [this survey](#).

  Student Edition pages and an assignable Digital Companion needed.

Instruction

Differentiation Beyond the Lesson

Big Idea: CC1: Variability

Today's Goal

- Language Goal:** Compare and contrast two data sets using the mean and standard deviation. (*Reading and Writing*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Additional Materials

Unit 7 Calculator Guide (optional), Show What You Know Sheet

Standards

Building On 6.SP.3 6.SP.5.c Addressing S-ID.2 S-ID.3 SMP.1 SMP.3 Building Toward S-ID.4

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.11 ELD.PI.9-10.3 ELD.PI.9-10.4

Review Vocabulary

- mean
- median

Instructional Routines

- Notice and Wonder
- Stronger and Clearer Each Time (MLR1)



Race Car

Lesson 10: Comparing Data Using Median and IQR

Let's compare measures of spread in skewed data sets.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Students use digital interactions to visualize data, compare dot plots, and develop conceptual understanding of IQR.

Instruction

Differentiation Beyond the Lesson

Big Idea: CC1: Variability

Today's Goals

- Understand that the interquartile range (IQR) is a measure of spread and calculate the IQR for a data set.
- Language Goal:** Compare and contrast skewed data sets using the median and IQR. (*Writing, Speaking, and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 3 to support math language acquisition for your students.

Standards

Building On 6.SP.3 6.SP.4 6.SP.5 Addressing S-ID.1 S-ID.2 S-ID.3 SMP.2 SMP.3 Building Toward S-ID.4

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.3 ELD.PI.9-10.5 ELD.PI.9-10.6 ELD.PI.9-10.8

Review Vocabulary

- interquartile range (IQR)

Instructional Routines

- Decide and Defend



Far Out

Lesson 11: Identifying Outliers

Let's determine whether or not a data point is an outlier, and consider its effect on the mean and median.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Students use the Desmos Graphing Calculator to identify outliers in data sets.

Instruction

Differentiation Beyond the Lesson

Big Idea: CC1: Variability

Today's Goals

- Determine whether or not a data point is an **outlier**.
- Language Goal:** Explain how outliers impact the mean or median of a data set. (*Speaking and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Additional Materials

Unit 7 Calculator Guide (optional)

Standards

Building On 6.SP.3 6.SP.5.c Addressing S-ID.3 SMP.2 SMP.3 SMP.5 Building Toward S-ID.2

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.12 ELD.PI.9-10.3 ELD.PI.9-10.6

Vocabulary

- outlier

Instructional Routines

- Compare and Connect (MLR7)
- Decide and Defend



Dynamic Decades

Lesson 12: Comparing Data Using Measures of Center and Spread

Let's use measures of center and spread to describe how aspects of the United States have changed over time.

Let us know what you thought about this lesson by filling out [this survey](#).



Student Edition pages and an assignable Digital Companion needed.

Instruction

Differentiation Beyond the Lesson

Big Idea: CC1: Variability

Today's Goal

- Language Goal:** Compare and contrast two data sets using measures of center and spread that are appropriate for the shape of the data.
(Writing, Speaking, and Listening)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Additional Materials

poster paper, coloring tools, Unit 7 Calculator Guide (optional), Show What You Know Sheet

Standards

Building On 6.SP.3 6.SP.4 6.SP.5.c 6.SP.5.d Addressing S-ID.1 S-ID.2 S-ID.3 SMP.1 SMP.3 SMP.4 SMP.5 Building Toward S-ID.4

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.2 ELD.PI.9-10.3 ELD.PI.9-10.5 ELD.PI.9-10.6

Instructional Routines

- Compare and Connect (MLR7)
- Co-Craft Questions (MLR5)
- Notice and Wonder



M1.7 Practice Day 1

Students practice the concepts, skills, and strategies developed in Lessons 1–12. Consider using this Practice Day as preparation for the upcoming Sub-Unit Quiz.

Let us know what you thought about this lesson by filling out [this survey](#).



Student Edition pages and an assignable Digital Companion needed.

Big Idea: CC1: Variability

Materials

- Task cards (one per group)
- Student Edition
- Digital Companion
- Unit 7 Calculator Guide (optional)

Standards

Addressing

S-ID.1

S-ID.2

S-ID.3

SMP.2



M1.7 Sub-Unit Quiz



Learn about your students' understanding of the concepts and skills so far in this unit.
This assessment is intended for students to complete using a graphing calculator.

Let us know what you thought about this lesson by filling out [this survey](#).

Instruction

Differentiation Beyond the Lesson

Big Idea: CC1: Variability

Standards

Assessing

S-ID.1

S-ID.2

S-ID.3

SMP.3

SMP.6



Correlation Coefficient

Lesson 13: Introduction to the Correlation Coefficient

Let's learn about the correlation coefficient (r -value) as a way to measure the strength and direction of a linear relationship.

 This activity contains a Polygraph. [Learn how to play.](#)

 Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Students use the Desmos Graphing Calculator to calculate the correlation coefficient and use digital interactions to make sense of what an r -value can tell you about the strength and direction of a linear association.

Instruction

Differentiation Beyond the Lesson

Big Ideas: CC1: Comparing Models, Correlation and Causation

Today's Goals

- Use technology to calculate the *correlation coefficient* of data on a *scatter plot*.
- **Language Goal:** Describe the strength and direction of a linear association between two variables on a scatter plot using the correlation coefficient. (*Speaking and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 2 to support math language acquisition for your students.

Additional Vocabulary

direction, strength

Additional Materials

Unit 7 Calculator Guide (optional)

Standards

Building On 8.SP.1 Addressing S-ID.8 SMP.1 SMP.6 SMP.7 Building Toward S-ID.6a

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.11 ELD.PI.9-10.12 ELD.PI.9-10.2 ELD.PI.9-10.4

Vocabulary

- correlation coefficient
- r -value
- scatter plot

Review Vocabulary

- linear association
- negative association
- positive association

Instructional Routines

- Collect and Display (MLR2)
- Critique, Correct, Clarify (MLR3)
- Notice and Wonder



How Hot Is It?

Lesson 14: Interpreting the Correlation Coefficient in Context

Let's use correlation coefficients to analyze relationships between income, tree cover, and average temperature.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Students can use digital interactions to explore and compare data.

Instruction

Differentiation Beyond the Lesson

Big Ideas: CC1: Comparing Models, Variability, Correlation and Causation

Instructional Routines

- Notice and Wonder

Today's Goals

- Use the correlation coefficient to describe the strength and direction of a linear association.
- **Language Goal:** Interpret a correlation coefficient in context. (*Writing, Speaking, and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 2 to support math language acquisition for your students.

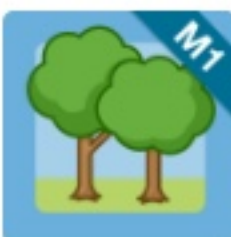
CA EP&C Principle 3

Standards

Building On 6.SP.1 Addressing N-Q.2 S-ID.6 S-ID.8 SMP.1 SMP.4 Building Toward S-ID.7 S-ID.8

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.3 ELD.PI.9-10.6



City Slopes

Lesson 15: Interpreting Slope and y -intercept in Context

Let's use a line of fit to describe the relationship between two variables and make predictions.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Students use digital manipulatives to create linear models to fit a line to data, make predictions, and view their predictions on a graph.

Instruction

Differentiation Beyond the Lesson

Big Ideas: CC1: Comparing Models, Variability

Today's Goals

- Estimate unknown values using a linear model for data on a scatter plot.
- Language Goal:** Interpret the slope and y -intercept for a linear model in context. (*Writing, Speaking, and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 2 to support math language acquisition for your students.

CA EP&C Principles 2, 4

Standards

Building On 8.SP.3 Addressing F-IF.4 F-LE.5 S-ID.6a S-ID.6c S-ID.7 SMP.1 SMP.4 SMP.6 Building Toward S-ID.6a

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.5 ELD.PI.9-10.6

Review Vocabulary

- line of fit
- slope
- slope-intercept form
- y -intercept

Instructional Routines

- Notice and Wonder
- Stronger and Clearer Each Time (MLR1)
- Think-Pair-Share



Residual Fruit

Lesson 16: Residuals and Residual Plots

Let's use residual plots to determine how well a line fits data.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Students explore dynamic residual plots and use them as a tool for assessing the fit of a line to data.

Instruction

Differentiation Beyond the Lesson

Big Ideas: CC1: Variability

Today's Goals

- Make connections between a *residual plots* and *residuals* on a graph.
- Use residuals and residual plots to understand what it means for a line to fit data well.
- **Language Goal:** Identify when a residual plot indicates a better or worse fit. (*Writing, Speaking, and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 2 to support math language acquisition for your students.

Standards

Building On 8.SP.2 8.SP.3 Addressing S-ID.6a S-ID.6b SMP.1 SMP.4 SMP.6 Building Toward S-ID.6a S-ID.6c

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.11 ELD.PI.9-10.12 ELD.PI.9-10.6

Vocabulary

- residual
- residual plot

Instructional Routines

- Notice and Wonder
- Collect and Display (MLR2)



Penguin Populations

Lesson 17: Using Technology to Generate the Line of Best Fit

Let's generate and analyze lines of best fit to explore how penguin populations have changed over time.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Students use the Desmos Graphing Calculator to create mathematical models.

Instruction

Differentiation Beyond the Lesson

Big Ideas: CC1: Modeling with Functions, Comparing Models , Variability

Today's Goals

- Use technology to generate the *line of best fit* for data on a scatter plot.
- Use the equation of a line of best fit to make predictions.
- **Language Goal:** Interpret the slope and y -intercept of the equation of a line of best fit in context. (*Speaking and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 2 to support math language acquisition for your students.

Additional Materials

Unit 7 Calculator Guide

Standards

Building On 8.SP.2 8.SP.3 Addressing F-LE.5 S-ID.6a S-ID.6c S-ID.7 S-ID.8 SMP.4 SMP.6 Building Toward S-ID.6a S-ID.9

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.12 ELD.PI.9-10.2 ELD.PI.9-10.5 ELD.PI.9-10.6

Vocabulary

- line of best fit

Instructional Routines

- Think-Pair-Share



Behind the Headlines

Lesson 18: Causation vs. Correlation

Let's consider the differences between correlation and causation.

Let us know what you thought about this lesson by filling out [this survey](#).



Student Edition pages and Presentation Screens support learning in this lesson.

Instruction

Differentiation Beyond the Lesson

Big Idea: CC1: Correlation and Causation

Today's Goals

- Determine if the relationship between two variables represents *correlation* or *causation*.
- Language Goal:** Analyze claims to determine if there is enough information to support causation. (*Reading, Writing, and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Additional Vocabulary

lurking variable

Additional Materials

Activity 2 Cards (one set per pair, pre-cut); Article Summary Sheet (one per student)

Standards

Building On 8.SP.1 Addressing S-ID.9 SMP.1 SMP.3 SMP.6 Building Toward S-ID.9

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.11 ELD.PI.9-10.3 ELD.PI.9-10.6

Vocabulary

- causation
- correlation

Instructional Routines

- Notice and Wonder
- Think-Pair-Share
- Discussion Supports (MLR8)



City Data

Lesson 19: Using Statistics to Analyze Relationships in Society

Let's apply what we've learned about analyzing two-variable data to explore relationships between variables in society.

Let us know what you thought about this lesson by filling out [this survey](#).

  Student Edition pages and an assignable Digital Companion needed.

Instruction Differentiation Beyond the Lesson

Big Ideas: CC1: Comparing Models, Correlation and Causation, Variability

Today's Goals

- Use statistical tools to explore relationships in real-world data.
- **Language Goal:** Analyze the relationship between two variables in context using scatter plots, lines of best fit, and correlation coefficients. (*Reading, Writing, Speaking, and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 2 to support math language acquisition for your students.

Additional Materials

Activity 1 Sheet (one per pair), Show What You Know Sheet, Unit 7 Calculator Guide (optional)

Standards

Building On 8.SP.1 8.SP.2 8.SP.3 Addressing S-ID.6 S-ID.6a S-ID.6c S-ID.7 S-ID.8 SMP.1 SMP.4 Building Toward S-ID.6a

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.11 ELD.PI.9-10.12 ELD.PI.9-10.5 ELD.PI.9-10.6

Review Vocabulary

- linear
- non-linear

Instructional Routines

- Notice and Wonder



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M1.7 Performance Task



Assign this summative assessment performance task at the end of the unit to evaluate students' proficiency with the concepts and skills addressed in the unit.

Let us know what you thought about this lesson by filling out [this survey](#).

Big Ideas: CC1: Variability, Correlation and Causation, Comparing Models

Standards

Assessing S-ID.1 S-ID.2 S-ID.3 S-ID.6b S-ID.8 S-ID.9 SMP.3 SMP.4



Unit Synthesis and Reflection



Here are six optional activities for students to engage in at the end of a unit to synthesize and/or reflect on their learning from the unit.

Notes:

- **Select one to two tasks** for students to complete before or after the End-Unit Assessment.
- Many tasks invite students to refer to the Student Goals and Glossary, which can be downloaded from the Unit Overview page at the beginning of each unit.

📧 Let us know what you thought about this lesson by filling out [this survey](#).

Learning Goals

- Be purposeful, strategic, and goal-oriented about learning.
- Connect different ideas from the unit and across units.
- Revisit ideas from the unit in order to make revisions and reflect on their growth.